

SECTION AAnswer **all** questions.

1. Addition of aqueous lead nitrate to aqueous potassium iodide causes a precipitate to form. Write an ionic equation for the formation of the precipitate, including state symbols. [1]

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2. Use the data shown to explain why silver chloride is insoluble in water. [2]

	Enthalpy / kJ mol^{-1}
Standard enthalpy of hydration for Ag^+	-464
Standard enthalpy of hydration for Cl^-	-364
Enthalpy of lattice formation of AgCl	-905

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3. A white crystalline solid gives a red colour in a flame test and addition of concentrated sulfuric acid to the solid gives purple fumes and a smell of rotten eggs. Name the ions present, giving your reasons. [2]

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4. Aluminium chloride forms Al_2Cl_6 dimers using coordinate bonding. Explain why the coordinate bonds form. [2]

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5. Addition of ammonia solution to aqueous copper(II) sulfate causes a pale blue precipitate to form followed by a coloured solution when excess ammonia solution is added.

Give the colour of the solution formed.

[1]

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6. The Haber process uses a heterogeneous catalyst.
State what is meant by the term 'heterogeneous' in this context.

[1]

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7. Hexagonal boron nitride is sometimes called white graphite. Give **one** difference between the physical properties of hexagonal boron nitride and graphite.

[1]

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